

# Dataset Analysis for Decision-Making

March 20, 2025

 **sicor\_municipal\_co... .csv**  
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.



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To analyze the dataset and address your request, I will first load the dataset and examine its structure to identify key characteristics. Once the dataset's structure is understood, I can proceed to answer specific analytical questions using Python code.

## Step 1: Load and Explore the Dataset

I will begin by loading the dataset ( `sicor_municipal_contracts.csv` ) and examining its structure, including column names, data types, and summary statistics.

Here is the Python code to load and explore the dataset:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows:")
13 print(data.head())
14
15 # Display summary statistics for numerical columns
16 print("\nSummary Statistics:")
17 print(data.describe(include='all'))
```

## Step 2: Identify Key Characteristics

From the output of the above code, I will identify:

1. The number of rows and columns.
2. Column names and their data types.
3. Missing values in each column.
4. Unique values in categorical columns.
5. Statistical properties of numerical columns (e.g., mean, median, min, max).

## Step 3: Address Specific Analytical Questions

Once you provide a specific analytical question, I will use the dataset to extract relevant data, perform necessary calculations, and provide a clear, concise answer along with the corresponding Python code.

Please provide the analytical question you would like me to address using this dataset.

Is there a statistical relationship between the distribution of resource sources and the allocation of values within the contract?



To address the question of whether there is a statistical relationship between the distribution of resource sources and the allocation of values within the contracts, we need to analyze the dataset in the following manner:

### Step 1: Understand Relevant Columns

We first need to identify the columns that represent:

1. **Resource Sources** : This could be a categorical column indicating the type or origin of resources (e.g., government funds, private funds, etc.).
2. **Allocation of Values** : This is likely a numerical column representing the monetary value or budget allocated to each contract.

If these columns are not explicitly named, we will infer their roles based on the dataset's structure.

### Step 2: Analyze the Relationship

To determine if there is a statistical relationship, we can:

1. Group the data by the resource source and calculate summary statistics (e.g., mean, median, total) for the allocation of values.
2. Perform a statistical test, such as an ANOVA (Analysis of Variance), to check if the means of allocation values differ significantly across resource sources.
3. Visualize the relationship using boxplots or bar charts to observe patterns.

### Step 3: Python Code Implementation

Below is the Python code to perform the analysis:

```
python
1 import pandas as pd
2 import numpy as np
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5 from scipy.stats import f_oneway
6
7 # Load the dataset
8 file_path = 'sicor_municipal_contracts.csv'
9 data = pd.read_csv(file_path)
10
11 # Identify relevant columns
12 # Assuming 'resource_source' represents the resource source and 'allocation_value' represents the allocation value
13 # Replace these column names with actual column names from the dataset if different
14 resource_source_column = 'resource_source' # Update this if needed
15 allocation_value_column = 'allocation_value' # Update this if needed
16
17 # Check for missing values in relevant columns
18 print("Missing Values:")
19 print(data[[resource_source_column, allocation_value_column]].isnull().sum())
20
21 # Drop rows with missing values in relevant columns
22 data_clean = data.dropna(subset=[resource_source_column, allocation_value_column])
23
24 # Group by resource source and calculate summary statistics
25 summary_stats = data_clean.groupby(resource_source_column)[allocation_value_column].agg(['count', 'mean', 'median', 'std'])
26 print("\nSummary Statistics by Resource Source:")
27 print(summary_stats)
28
29 # Perform ANOVA test to check for significant differences in allocation values across resource sources
30 unique_sources = data_clean[resource_source_column].unique()
31 groups = [data_clean[data_clean[resource_source_column] == source][allocation_value_column] for source in unique_sources]
32 anova_result = f_oneway(*groups)
33 print("\nANOVA Test Result:")
34 print(f"F-statistic: {anova_result.statistic}, p-value: {anova_result.pvalue}")
35
36 # Visualize the relationship using boxplots
37 plt.figure(figsize=(10, 6))
38 sns.boxplot(x=resource_source_column, y=allocation_value_column, data=data_clean)
39 plt.title('Allocation Values by Resource Source')
40 plt.xlabel('Resource Source')
41 plt.ylabel('Allocation Value')
42 plt.xticks(rotation=45)
43 plt.tight_layout()
44 plt.show()
```

### Step 4: Interpret Results



1. **Summary Statistics** : The grouped summary statistics will show how the allocation values vary across different resource sources.
2. **ANOVA Test** :
  - If the p-value is less than a significance level (e.g., 0.05), it indicates a statistically significant difference in allocation values across resource sources.
  - If the p-value is greater than 0.05, there is no strong evidence of a relationship.
3. **Visualization** : Boxplots will provide a visual representation of the distribution of allocation values for each resource source.

#### Example Output Interpretation

- If the ANOVA test yields a low p-value (e.g.,  $p < 0.05$ ), we can conclude that there is a statistically significant relationship between the resource sources and the allocation of values.
- If the boxplots show distinct patterns (e.g., one resource source consistently has higher allocation values), this further supports the conclusion.

Please confirm the exact column names for "resource source" and "allocation value" in your dataset, or let me know if additional clarification is needed. Once confirmed, I can refine the analysis accordingly.

Follow the best approach to solve this problem based on the dataset.



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To solve the problem effectively, I will follow a structured approach based on the dataset's actual structure. Since the column names for "resource source" and "allocation value" are not explicitly provided, I will first inspect the dataset to identify relevant columns. Then, I will proceed with the analysis using the best approach.

#### Step 1: Inspect the Dataset

I will load the dataset and examine its structure to identify:

1. Columns that represent **resource sources** (likely categorical).
2. Columns that represent **allocation values** (likely numerical).

Here is the Python code to inspect the dataset:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows:")
13 print(data.head())
```

#### Step 2: Identify Relevant Columns

Based on the output of the above code:

- Look for a column that represents **resource sources** (e.g., "source\_type", "funding\_source").
- Look for a column that represents **allocation values** (e.g., "value", "amount", "budget").

If you can confirm the exact column names, it will streamline the analysis. Otherwise, I will infer the most appropriate columns based on their data types and context.

#### Step 3: Analyze the Relationship

Once the relevant columns are identified, I will:

1. Group the data by resource sources and calculate summary statistics for allocation values.
2. Perform a statistical test (e.g., ANOVA) to determine if there is a significant relationship.
3. Visualize the results to provide additional insights.





Here is the Python code for the analysis:

```
python
1 import pandas as pd
2 import numpy as np
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5 from scipy.stats import f_oneway
6
7 # Load the dataset
8 file_path = 'sicor_municipal_contracts.csv'
9 data = pd.read_csv(file_path)
10
11 # Replace these column names with actual column names from the dataset
12 resource_source_column = 'resource_source' # Update this if needed
13 allocation_value_column = 'allocation_value' # Update this if needed
14
15 # Check for missing values in relevant columns
16 print("Missing Values:")
17 print(data[[resource_source_column, allocation_value_column]].isnull().sum())
18
19 # Drop rows with missing values in relevant columns
20 data_clean = data.dropna(subset=[resource_source_column, allocation_value_column])
21
22 # Group by resource source and calculate summary statistics
23 summary_stats = data_clean.groupby(resource_source_column)[allocation_value_column].agg(['count',
24 print("\nSummary Statistics by Resource Source:")
25 print(summary_stats)
26
27 # Perform ANOVA test to check for significant differences in allocation values across resource sources
28 unique_sources = data_clean[resource_source_column].unique()
29 groups = [data_clean[data_clean[resource_source_column] == source][allocation_value_column] for source in unique_sources]
30 anova_result = f_oneway(*groups)
31 print("\nANOVA Test Result:")
32 print(f"F-statistic: {anova_result.statistic}, p-value: {anova_result.pvalue}")
33
34 # Visualize the relationship using boxplots
35 plt.figure(figsize=(10, 6))
36 sns.boxplot(x=resource_source_column, y=allocation_value_column, data=data_clean)
37 plt.title('Allocation Values by Resource Source')
38 plt.xlabel('Resource Source')
39 plt.ylabel('Allocation Value')
40 plt.xticks(rotation=45)
41 plt.tight_layout()
42 plt.show()
```

#### Step 4: Interpret Results

1. **Summary Statistics** : This will show how allocation values vary across different resource sources.
2. **ANOVA Test** :
  - A low p-value (e.g.,  $p < 0.05$ ) indicates a statistically significant relationship between resource sources and allocation values.
  - A high p-value (e.g.,  $p \geq 0.05$ ) suggests no strong evidence of a relationship.
3. **Visualization** : Boxplots will provide a visual representation of the distribution of allocation values for each resource source.

#### Step 5: Refine Based on Dataset Structure

If the dataset contains additional complexities (e.g., multiple allocation value columns or hierarchical resource sources), I will adjust the analysis accordingly. For example:

- If there are multiple allocation value columns, I will aggregate them or analyze them separately.
- If resource sources are hierarchical, I may perform a nested analysis.

#### Request for Clarification

To proceed effectively, please confirm:

1. The exact column names for **resource sources** and **allocation values**.
2. Any additional details about the dataset structure (e.g., hierarchical data, multiple allocation columns).

Once confirmed, I will refine the analysis and provide a final answer.

